

NON-REDUNDANCY OF THE CASAS–ALVERO CONDITIONS: AN ELEMENTARY LEMMA AND ITS CONSEQUENCES

ABSTRACT. Let $f = x^d + a_1x^{d-1} + \cdots + a_{d-1}x$ be generic and let $R_i = \text{Res}(f, H_i(f))$ be the resultant of f with its i -th Hasse derivative. Schaub and Spivakovsky proved that $R_i \notin \sqrt{(R_j : j \neq i)}$ for $i \in \{d-3, d-2, d-1\}$. We show that this holds for *every* $i \in \{1, \dots, d-1\}$, with an elementary proof based on explicit witnesses: on the lines $t \mapsto x^d + tx^i$ (and $t \mapsto x^d - tx^{d-1}$) all the resultants R_j with $j \neq i$ vanish and $R_i = \pm \left(\binom{d}{i} - 1\right) t^{d-i}$. We also discuss the consequences: none of the $d-1$ Casas–Alvero conditions is redundant; the conjecture is equivalent to R_i not vanishing identically on *any* component of $V(R_j : j \neq i)$, and our lemma establishes this for *one*; and every membership certificate $a_k^N \in (R_1, \dots, R_{d-1})$ is strongly constrained along those lines.

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1. SETTING

We follow the notation of Schaub–Spivakovsky [7]. Let K be a field of characteristic 0, let $d \geq 3$, and let

$$f = x^d + a_1x^{d-1} + \cdots + a_{d-1}x \in K[a_1, \dots, a_{d-1}][x],$$

that is, with $a_0 = 1$ and $a_d = 0$ (after a translation one may always assume that 0 is a root of f). For $1 \leq i \leq d-1$ let

$$H_i(f) = \sum_{k=0}^{d-i} \binom{d-k}{i} a_k x^{d-k-i}$$

be the i -th Hasse derivative, $H_i(f) = f^{(i)}/i!$, and set

$$R_i := \text{Res}(f, H_i(f)) \in K[a_1, \dots, a_{d-1}], \quad J := (R_1, \dots, R_{d-1}), \quad J_i := (R_j : j \neq i).$$

A polynomial f has a non-constant common factor with $H_i(f)$ if and only if $R_i(f) = 0$.

Conjecture 1.1 (Casas–Alvero). *If f has a non-constant common factor with every $H_i(f)$, $1 \leq i \leq d-1$, then $f = x^d$. Equivalently, $V(J) = \{0\}$; equivalently, $a_k^N \in J$ for every k and some N .*

Since the R_j have integer coefficients, the validity of the last formulation does not depend on K (in characteristic 0), and for that reason everything below is over $\mathbb{Q}$. Schaub and Spivakovsky also observe [7, Rmk. 4] that the conjecture is equivalent to $R_1, \dots, R_{d-1}$ being a regular sequence in $K[a_1, \dots, a_{d-1}]$, in any order.

Quasi-homogeneity. With the weights $\deg a_k = k$, each R_j is quasi-homogeneous: the substitution $a_k \mapsto \lambda^k a_k$ corresponds to $f(x) \mapsto \lambda^d f(x/\lambda)$, which rescales the roots of f and of $H_j(f)$ by λ . Consequently $V(J)$ and $V(J_i)$ are cones, stable under the action of $\overline{K}^*$.

2. THE RESULT OF SCHAUB-SPIVAKOVSKY

Theorem 2.1 ([7, Theorem 5]). *If $i \in \{d-3, d-2, d-1\}$ then $R_i \notin \sqrt{(R_1, \dots, \check{R}_i, \dots, R_{d-1})}$.*

This is a partial result “in the direction” of the conjecture: if the conjecture holds then $R_i \notin \sqrt{J_i}$ for *every* i , because if R_i were in the radical of the others we would have $V(J_i) = V(J)$, and the height of J would be at most $d-2$.

The proof in [7] is indirect. Denying $R_i \in \sqrt{J_i}$ amounts to exhibiting f with $R_j(f) = 0$ for $j \neq i$ and $R_i(f) \neq 0$. Draisma and de Jong [3] construct, for each i , real polynomials satisfying all the conditions $j \neq i$ which are not powers of a linear polynomial (*almost counterexamples of level i*). But “not being a power of a linear polynomial” does not imply $R_i(f) \neq 0$ without already knowing the conjecture. To rule out those polynomials being genuine counterexamples, [7] combines arguments on the location of real roots (Rolle) with the fact that a counterexample has at least five distinct roots [2]; those arguments only close for the three highest indices.

3. THE LEMMA: ALL INDICES

The idea is to use *degenerate* witnesses: a root 0 of multiplicity i is shared by all the derivatives except the i -th, and checking $R_i \neq 0$ becomes a direct computation.

Proposition 3.1 (values on the witness lines). *For $t \in K$ let*

$$f_{i,t} := x^d + tx^i = x^i(x^{d-i} + t) \quad (1 \leq i \leq d-2), \quad f_{d-1,t} := x^d - tx^{d-1} = x^{d-1}(x - t).$$

Then, for every $i \in \{1, \dots, d-1\}$,

$$R_j(f_{i,t}) = 0 \quad (j \neq i), \quad R_i(f_{i,t}) = \pm \left(\binom{d}{i} - 1 \right)^{d-i} t^d.$$

Proof. We use Poisson’s formula $\text{Res}(f, g) = \pm \text{lc}(g)^{\deg f} \prod_{g(\eta)=0} f(\eta)$.

Case $1 \leq i \leq d-2$. Here $H_j(f_{i,t}) = \binom{d}{j}x^{d-j} + t\binom{i}{j}x^{i-j}$.

- If $j < i$, both summands vanish at 0, and $f_{i,t}(0) = 0$; hence $R_j = 0$.
- If $j > i$, then $\binom{i}{j} = 0$ and $H_j = \binom{d}{j}x^{d-j}$ with $d-j \geq 1$; it shares the root 0, hence $R_j = 0$.
- If $j = i$, set $C = \binom{d}{i}$ and $m = d-i$. Then $H_i = Cx^m + t$, whose roots $\eta_1, \dots, \eta_m$ satisfy $\eta^m = -t/C$ and $\prod \eta_\ell = \pm t/C$. At those roots

$$f_{i,t}(\eta) = \eta^i(\eta^m + t) = \eta^i t \left(1 - \frac{1}{C}\right),$$

so that

$$R_i = \pm C^d \left(\prod_{\ell} \eta_{\ell} \right)^i t^m \left(\frac{C-1}{C} \right)^m = \pm C^{i+m} \frac{t^i}{C^i} t^m \frac{(C-1)^m}{C^m} = \pm (C-1)^{d-i} t^d.$$

Case $i = d-1$. Here $H_j(f_{d-1,t}) = \binom{d}{j}x^{d-j} - t\binom{d-1}{j}x^{d-1-j}$. If $j \leq d-2$ both exponents are ≥ 1 , so H_j shares the root 0 with f and $R_j = 0$. If $j = d-1$, then $H_{d-1} = dx - t$ has root t/d and $f_{d-1,t}(t/d) = (t/d)^{d-1}(t/d - t) = -(d-1)t^d/d^d$, whence $R_{d-1} = \pm d^d \cdot (d-1)t^d/d^d = \pm (d-1)t^d$, which is the formula with $\binom{d}{d-1} - 1 = d-1$. $\square$

Theorem 3.2. *For every $d \geq 3$ and every $i \in \{1, \dots, d-1\}$,*

$$R_i \notin \sqrt{(R_1, \dots, \check{R}_i, \dots, R_{d-1})} \quad \text{in } K[a_1, \dots, a_{d-1}].$$

Proof. Suppose $R_i^N = \sum_{j \neq i} c_j R_j$ with $c_j \in K[a]$. Evaluating at the K -rational point corresponding to $f_{i,1}$, Proposition 3.1 gives $0 \neq (\pm((\binom{d}{i} - 1)^{d-i})^N = 0$, since $\binom{d}{i} \geq d \geq 3$ for $1 \leq i \leq d-1$. Contradiction. $\square$

Remark 3.3. The proof does not use the Nullstellensatz: only the trivial direction “if R_i lies in the radical then it vanishes at every zero of J_i ”. It uses neither real roots nor the conjecture. In coordinates, the witness of index $i \leq d-2$ is the point with $a_{d-i} = 1$ and all other coordinates zero; the one of index $d-1$ is $a_1 = -1$. In both cases the coordinate that moves is a_{d-i} .

Remark 3.4. Theorem 3.2 has been formally verified in Lean 4 with Mathlib, for every d and every i , over an arbitrary field of characteristic zero: see the theorems `exists_witness` and `exists_witness_resultant` in the accompanying development.

4. CONSEQUENCES

4.1. No condition is redundant.

Corollary 4.1. *For each i , the $d-2$ conditions “ f shares a root with $H_j(f)$, $j \neq i$ ” do not imply the i -th one, not even at the level of radicals. More precisely, the line $L_i := \{f_{i,t} : t \in K\}$ is contained in $V(J_i)$ and $L_i \cap V(R_i) = \{0\}$.*

Note the difference with the almost counterexamples of [3]: the polynomials $f_{i,1}$ not only satisfy the conditions $j \neq i$, they *verifiably violate* the i -th one.

4.2. Where the difficulty lies: one component against all.

Lemma 4.2. *Fix i . Conjecture 1.1 holds in degree d if and only if every irreducible component C of $V_{\overline{K}}(J_i)$ satisfies $\dim C = 1$ and $R_i|_C \not\equiv 0$.*

Proof. ($\Leftarrow$) Each component C is stable under the action of $\overline{K}^*$ (the group is connected) and contains 0 (the limit $\lambda \rightarrow 0$). If $\dim C = 1$ and $p \in C \setminus \{0\}$, the orbit of p is infinite (the weights are positive), so its closure is an irreducible closed set of dimension 1 contained in C , hence equal to C . Since R_i is quasi-homogeneous, $R_i(\lambda \cdot p) = \lambda^w R_i(p)$; if $R_i(p) = 0$ for some $p \neq 0$ in C then R_i vanishes on the whole orbit, hence on C , contrary to hypothesis. Therefore $C \cap V(R_i) = \{0\}$ for every component, and $V(J) = V(J_i) \cap V(R_i) = \{0\}$.

($\Rightarrow$) Suppose $V(J) = \{0\}$. By Krull’s principal ideal theorem, every component of $V(J_i)$ (defined by $d-2$ equations in $d-1$ variables) has dimension ≥ 1 . If some C had dimension ≥ 2 , then $C \cap V(R_i)$ would be nonempty (it contains 0) of dimension ≥ 1 , contained in $V(J)$, absurd. And if $R_i \equiv 0$ on some C , then $C \subseteq V(J)$, also absurd. $\square$

Theorem 3.2 says that the component of $V(J_i)$ containing L_i satisfies $R_i|_C \not\equiv 0$. The conjecture asks for the same on *every* component (and for all of them to be curves). In particular, if the conjecture holds then L_i is an irreducible component of $V(J_i)$. The distance between “one component” and “all of them” is exactly the difficulty of Casas–Alvero.

4.3. Constraints on membership certificates. The membership formulation of the conjecture asks, for each k , for a certificate

$$a_k^N = \sum_{j=1}^{d-1} c_j R_j, \quad c_j \in \mathbb{Q}[a_1, \dots, a_{d-1}]. \quad (1)$$

Let $\varphi_i : \mathbb{Q}[a] \rightarrow \mathbb{Q}[t]$ be the restriction to the line L_i , that is, $a_{d-i} \mapsto t$ (or $a_1 \mapsto -t$ if $i = d-1$) and all other coordinates $\mapsto 0$. Each coordinate a_k moves on exactly one of the lines, the one of index $i = d - k$.

Corollary 4.3. *Every certificate (1) satisfies:*

- (1) $N \geq d$;
- (2) $\varphi_{d-k}(c_{d-k}) = \pm t^{N-d} / ((\binom{d}{k} - 1)^k)$; in particular, the certificate for a_k^N must use R_{d-k} ;
- (3) $\varphi_i(c_i) = 0$ for every $i \neq d - k$.

Proof. Apply φ_i to (1). By Proposition 3.1 only the term $j = i$ survives: $\varphi_i(a_k)^N = \varphi_i(c_i) \cdot (\pm (\binom{d}{i} - 1)^{d-i} t^d)$. If $i = d - k$, the left-hand side is $\pm t^N$, which forces $N \geq d$ and the value of $\varphi_{d-k}(c_{d-k})$, using $\binom{d}{d-k} = \binom{d}{k}$. If $i \neq d - k$, the left-hand side is 0 and, since $(\binom{d}{i} - 1)^{d-i} t^d \neq 0$ in $\mathbb{Q}[t]$, we get $\varphi_i(c_i) = 0$. $\square$

These conditions are the natural starting point for a search for explicit certificates over $\mathbb{Q}$: they fix the behaviour of each coefficient c_j on $d - 1$ explicit lines.

4.4. Relation to Ghosh’s proof. Ghosh [5] announced a complete proof of the conjecture (Koszul homology plus Brouwer degree arguments), revised in March 2026 and, as far as we could see, not published in a journal; an August 2026 preprint still treats it as open. If Ghosh’s proof is correct, it implies Theorem 3.2, and what this note contributes is an elementary and independent proof. Should it contain an error, Theorem 3.2 stands as an unconditional partial result extending Theorem 2.1.

5. COMPUTATIONAL VERIFICATION

Everything was verified with SageMath 10.7 in the exact setting of [7] (Hasse derivatives, f unnormalised except for $a_d = 0$).

Statement	Verification
Prop. 3.1	all 284 resultants $R_j(f_{i,t})$, $3 \leq d \leq 10$, all i, j , with t symbolic: no failures.
Theorem 3.2, by evaluation	$3 \leq d \leq 9$, all i .
Theorem 3.2, by Gröbner bases	a radical test independent of the witnesses (Rabinowitsch’s trick, $1 \in J_i + (1 - yR_i)$): $d = 3, 4, 5$, all i , with outcome “not a member”.
Theorem 3.2, in Lean 4	all d and all i , over any field of characteristic zero, with no <code>sorry</code> .

6. CAVEATS

- We checked the statement of Theorem 2.1 in version 7 of [7] (February 2025). It is striking that the general case is not observed there; we do not rule out that it appears elsewhere, or that the interest of [7] lay in the real-root method rather than in the statement.
- This is not a deep result: it is a lemma. Its main use is Corollary 4.3, which constrains membership certificates over $\mathbb{Q}$, and Lemma 4.2, which locates the difficulty precisely.

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